

WASP-43b

R-2007

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_200125 · created 2026-08-07T16:43:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458873.8716 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.056
Transit depth (Rp/Rs)^2	3.74 +/- 2.17 [%]
Orbital Inclination (inc)	77.64 +/- 2.9
Airmass coefficient 1 (a1)	0.177 +/- 0.085
Airmass coefficient 2 (a2)	1.26 +/- 1.15
Scatter in the residuals of the lightcurve fit is	49.02 %
Best Comparison Star	None
Optimal Aperture	4.37
Optimal Annulus	12.57
Transit Duration (day)	0.027 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.056 vs 0.1594 (deviation 0.6σ)
Duration fit vs published	0.027 d vs 0.0512 d (deviation 1.21σ)
Mid-time O–C	-3.6 min
Depth SNR	3.4
Residual scatter	49.02 %

Light curve

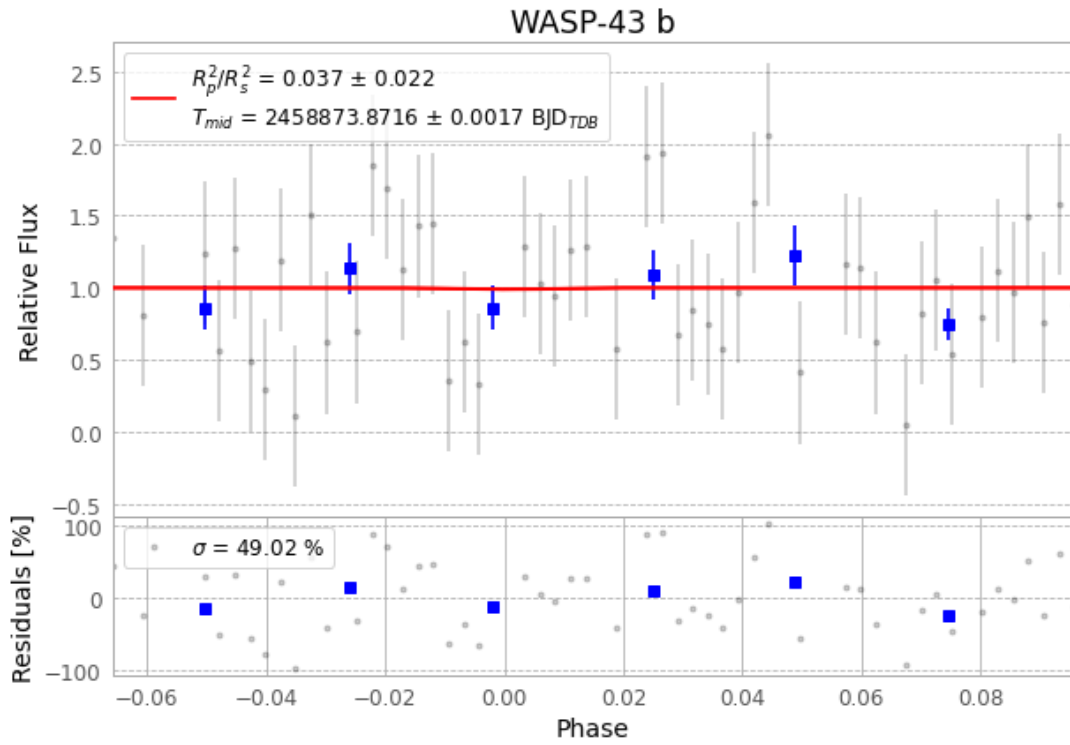


Figure 1 — FinalLightCurve_WASP-43 b_2020-01-25.png (SHA-256 b946391f4d88f461...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [243, 159], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6e655cb577902fa9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43200125073012.FITS → WASP-43200125103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200125.log (db8e48f886ef...), FinalLightCurve_WASP-43 b_2020-01-25.png (b946391f4d88...), FinalLightCurve_WASP-43 b_2020-01-25.csv (d5e9843b9d9d...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:43Z.